

10 A As the Earth is taken to be a uniform sphere, the gravitational potential at any point on the Earth's surface is the same (as $\phi = -\frac{GM}{R}$, where M is the mass of the Earth and R is the radius of the Earth), hence the gravitational potential energy of the Earth-rocket system is the same whether the rocket is near Equator or near the North Pole. As such the escape speed is the same whether the launch is near the Equator or near the North Pole.

11 B Those who chose C or D did not know that gravitational potential energy is negative. Those who